SAI ARUN GUNUGANTI

Senior Software Engineer · Lead Developer | Frontend Platform, Micro-Frontends & Agentic Systems

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SUMMARY

Senior Software Engineer and Lead Developer with 10+ years building large-scale web platforms for retail, banking, and healthcare (Lowe's, Apple, Cisco, McKesson, M&T Bank). At Lowe's I lead a team of 5 and own the checkout-critical Customer Lookup micro-frontend for the myRedVest POS platform, which I built from scratch and which 20+ teams integrate into their own applications (Lowe's serves ~16M customer transactions a week across 1,700+ stores). I specialize in micro-frontend architecture, cross-team platform contracts, test automation for Electron desktop apps, and agentic systems, including a live AI-agent authorization product (Sealgraf) with a published technical paper.

CORE SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, HTML5, CSS3
- **Frontend & Platform:** React, Redux / Redux-Saga, React Query, Vue.js, Angular, Webpack Module Federation (Micro-Frontends), design systems
- **Backend & APIs:** Node.js, Backend-for-Frontend (Micronaut / Spring Boot), REST APIs, LoopBack
- **Testing & Automation:** Playwright (Electron native apps), Jest, React Testing Library, Mock Service Worker
- **AI / Agentic:** Agentic system architecture (Python, Google agent SDK), MCP, autonomous test spec generation
- **Delivery & Observability:** GCP, CI/CD (Jenkins, Azure DevOps), ELK (Elasticsearch / Kibana), Grafana, FullStory
- **Leadership:** Technical leadership, mentoring and code review, design docs, engineering standards, cross-team integration

PROFESSIONAL EXPERIENCE

Lowe's Companies Inc. — Senior Software Engineer, Lead Developer (myRedVest Showroom POS) Apr 2022 – Present

- Built the **Customer Lookup Micro-Frontend** from scratch using Webpack Module Federation, custom events, and React Query; it is **integrated by 20+ teams** into their own applications across Lowe's. This checkout-critical capability covers customer add/edit, addresses, preferred contact method, tax-exempt and military discount, invoice/estimate recall, and adding customers to cart.
- Lead Developer for a **team of 5 engineers** (3 onshore, 2 offshore) and single point of contact for a core area of the checkout transaction flow. I mentor engineers and own code review for the team.
- Authored the design docs and integration conventions (event contracts, data requirements, usage patterns) that consuming teams follow, and onboarded those teams onto the Customer MFE with documentation and hands-on integration support.
- Designed and built the end-to-end **Playwright automation suite** for the myRedVest POS desktop application (Electron), run in UAT ahead of production deploys. It **caught critical failures in key POS transaction flows before release**, and I escalated them to the owning teams so catastrophic checkout failures were fixed before reaching stores.
- Set testing and front-end standards for the team, covering automation coverage expectations for checkout and customer-lookup flows and reusable patterns contributed to the myRedVest design system with the UX team.
- Designed an **agentic architecture for POS guided selling**, guiding store associates to the next step to complete the transaction journey, using Python and Lowe's internal version of Google's agent SDK. I also designed a POC for an autonomous test spec generator for POS testing.
- Integrated the checkout micro-frontends with common-layer BFF services (Micronaut / Spring Boot) as a consuming client, defining frontend data requirements and API contracts.
- First point of contact for production issues and change coordinator through every stage to production: creating CRQs, validating deployments, and presenting in CAB meetings for approvals up to Director/CIO level. Tuned application performance (Webpack/gzip configuration, FullStory, Kibana, Grafana, ELK instrumentation).

Atos Syntel Inc. — Consultant Nov 2020 – Apr 2022

- **McKesson (AIMS):** full-stack development with a Vue.js front-end focus: reusable Vue single-file components, Vuex state management, and Element UI. Resolved Okta Verify issues on the JavaScript side.
- **Baylor Scott & White Health:** React applications using React Query and Mock Service Worker, plus reusable React components for the bswhealth.com Sitecore migration and a .NET-based clinical trials application.

SLK America Inc. — Senior Software Engineer Sep 2018 – Nov 2020

- **First Horizon Bank:** front-end developer for the Loan Operations BPM tool (Angular 6 / TypeScript with a Node.js LoopBack API backend). Built reusable Angular component libraries (forms, designer, eTime).
- **M&T Bank:** digital banking products in React: e-statements, EZ-Money (P2P), Alexa banking integration, and a feature-flag dashboard. Proposed the Git flow for the TFS to Azure DevOps migration and implemented WCAG/ADA accessibility standards.

Valley Soft Systems Inc. — Software Engineer Sep 2017 – Aug 2018

- **Apple (Talent Point):** full-stack Meteor / React / Node.js / MongoDB work on Apple's careers recruitment platform.
- **Cisco (DNA Center):** React/Redux product UI and the shared Cisco-DNA-react-components library (data table, site selector, scheduler).

Alliance Information Technology Inc. — Software Engineer / Front-End Developer Aug 2016 – Sep 2017 · Jun 2014 – Aug 2015 (India)

- **Symantec (Rev-Pro):** React/TypeScript executive financial dashboard used for C-level operational reporting. Earlier front-end development (JavaScript/jQuery) for regional vendor dashboards in India.

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- **Sealgraf** (live product, sealgraf.com): designed and built the authorization layer for AI-agent spending. Owner-set limits are checked before money moves, approval happens on a surface the agent cannot fake, and every transaction gets a sealed, verifiable receipt. Built integration packages for **6 agent frameworks** (Dify, CrewAI, LangChain, Vercel AI SDK, n8n, Composio; 302 tests), each live-tested end-to-end. Authored the technical paper *"Approval You Can Verify"* (sealgraf.com/paper).
- **Doorway** (live demo): built an EAP-access agent with a state-machine orchestrator, rules engine, and hash-chained ledger. It verifies eligibility, explains benefits with citations, supports steerable matching and approval-gated booking, and keeps employer views aggregate-only. 44 tests
- **ListProof** (live demo): built a rental-listing verification pipeline covering ownership cross-checks, photo provenance, and price vs. market, with evidence-backed verdicts and a fee ledger where any change invalidates approval. 58 tests
- **Sector products** (live public demos): Dossier (travel recovery), JobProof (local-services evidence), Reclaim (rewards disputes), Refile (claims appeals), Destorm (incident triage), and ModSeal (moderation record). All enforce rules in code with approval gates and sealed receipts. 276 tests (378 across all eight sector products)
- **Reify** (open source, github.com/saiarung27/reify): built a local-first memory engine for AI agents with a bitemporal graph, hybrid retrieval, and MCP / REST / CLI surfaces. 442 tests
- **AgentFoundry**: an assurance platform for AI agents, with a detection engine, financial guardrails, and an MCP trust-layer proxy issuing allow / warn / block verdicts. ~195 tests

EDUCATION

- M.S., Computer Information Systems & IT — University of Central Missouri, Dec 2016
- B.Tech, Mechanical Engineering — Anurag Engineering College, 2014